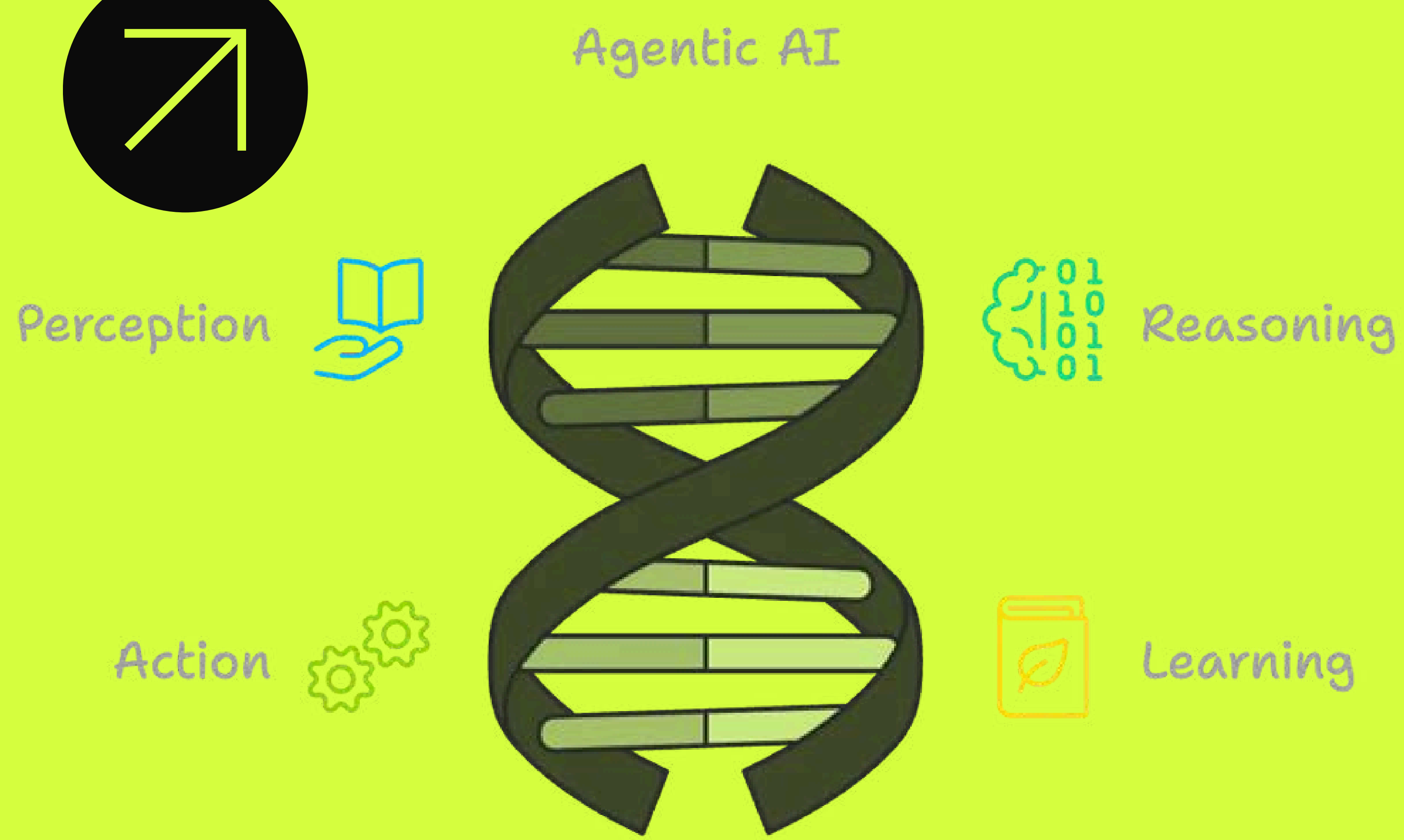
Md Amaan Nehru

Cozmo AI

PRODUCT

ASSIGNMENT



Presentation for a $50M investment thesis in vertical AI agents over two years.
Elements: Market sizing, adoption mapping across industries: BFSI, healthcare, and retail. A comparison of voice & non-voice opportunities. Include a competitive landscape analysis.
Provide data-driven recommendations focusing on one or two AI agent categories to prioritize
Include details on entry strategies, defensibility, and the ideal profiles of companies to target.

AI AGENTS HISTORY

Agentic Roadmap

From scripted responders to autonomous workers

2024

TRADITIONAL CHATBOTS USED FIXED DECISION TREES, WHILE MODERN LLM CHATBOTS OFFERED FLEXIBLE LANGUAGE UNDERSTANDING BUT WERE STILL REACTIVE. AI AGENTS HAVE EVOLVED INTO PROACTIVE SYSTEMS THAT CAN INDEPENDENTLY PLAN WORKFLOWS, RETRIEVE INFORMATION, AND EXECUTE COMPLEX TASKS, TRANSFORMING FROM BASIC Q&A INTERFACES INTO AUTONOMOUS DIGITAL COWORKERS.

/1

LLM Powered Chatbots

A conversational interface that uses a large language model to generate responses. It's relatively simple

/2

Agentic RAG

Works collaboratively, suggesting code, text, or actions, while the human makes final decisions. It emphasizes intelligent, adaptive systems that solve complex problems in real-time.

/3

AI AGENT

A system that uses sensors to perceive its environment, processes that information, and acts through actuators to achieve specific goals, functioning similarly to humans in a programmed manner.

The Rise

Attention, Memory & Context

2024

THE RISE OF AI AGENTS.

/1

Reasoning & Memory

The rise of AI agents is fueled by the advanced reasoning abilities of modern LLMs, enabling them to tackle complex problems, plan workflows, and adapt strategies. Persistent memory systems enhance this reasoning, allowing agents to maintain short-term context, store long-term preferences, and build procedural memory, facilitating learning from past interactions and personalizing behavior.

/2

Multi-Agent Orchestration & Function Calling

Agents now possess effective "hands" through reliable tool use and function calling, allowing them to chain multiple tools and manage responses or errors. This is enhanced by multi-agent orchestration, where specialized agents work in teams on tasks such as research, coding, or analysis, often in hierarchical structures, to efficiently achieve complex goals.

/3

Workflow Autonomy

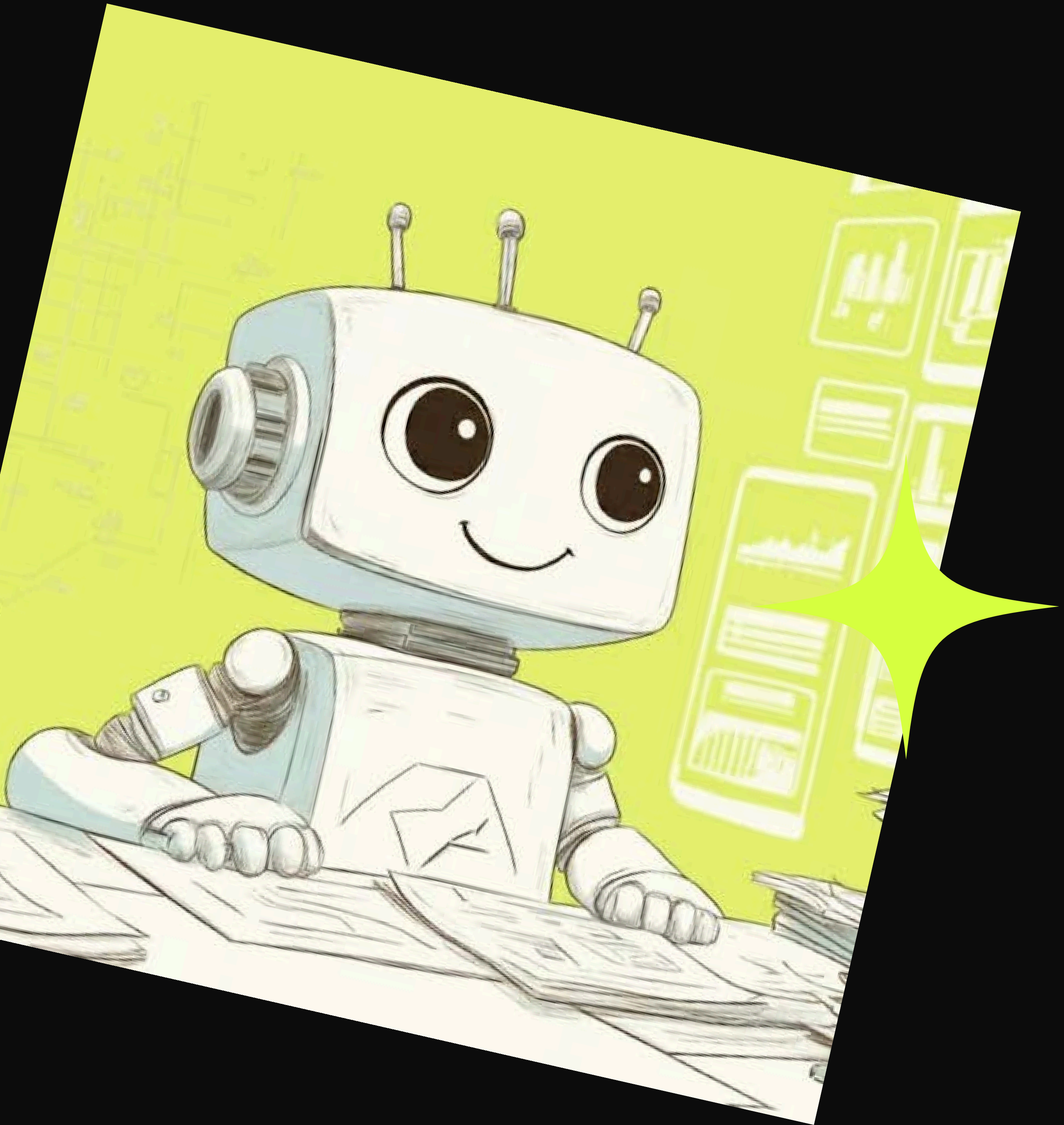
The advancements lead to true workflow autonomy, enabling agents to function independently for extended periods. They can self-correct errors and make decisions within set parameters, seeking human input only when necessary, transforming from simple tools into autonomous partners.

Opportunity Area

Voice vs Non voice

2024

CUSTOMER MOAT / THE VALUE ENGINE



Autonomous voice agents focus on real-time customer engagement to build a competitive advantage by providing a superior 24/7 experience that fosters brand loyalty. Their effectiveness is seen in high-value workflows such as outbound sales (increasing revenue by 2.5-3x), customer support (achieving up to 80% autonomous resolution), and collections or renewals (boosting retention by 25%). The main value lies in direct revenue generation and improved customer retention.

38.1% CAGR | Voice | $39.2B by 2030

Non-voice agents focus on autonomous back-office tasks, data processing, and workflow execution to enhance operational efficiency and reduce risk. They achieve significant returns in accuracy-critical processes, such as:

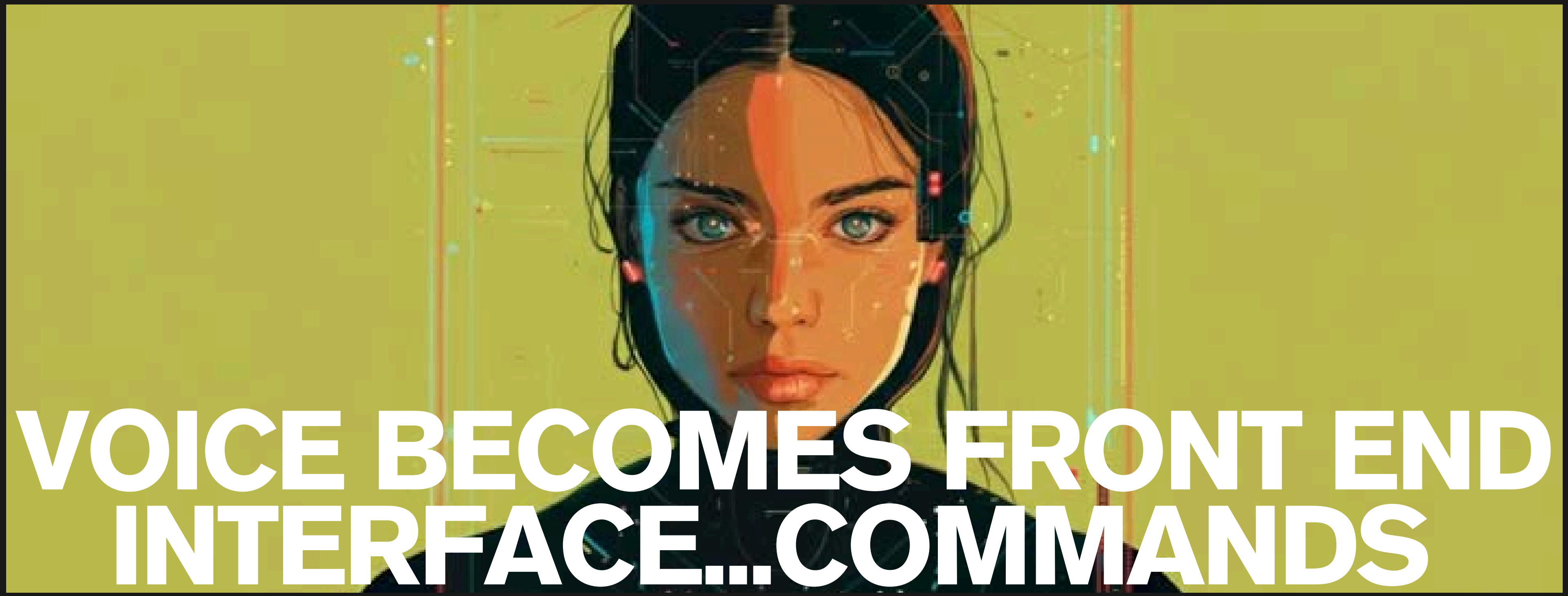
- Invoice processing (95% accuracy, high ROI)
- Contract review (75% time reduction)
- Real-time compliance monitoring

Their value comes from cost reduction and capturing institutional intellectual property within automated systems.

42.6% CAGR | Non-voice | $96.7B by 2030

Convergence Voice & Non voice

2024



VOICE BECOMES FRONT END INTERFACE...COMMANDS NON VOICE BACK END AGENT TEAMS

The Short-Term Pattern

Voice = Quick wins, executive sponsorship, proven ROI (call deflection, conversion rates), C suite visibility, direct impact reflected on P&L, infrastructure ready, less organisational resistance

Non-Voice = Longer sales cycles, IT/legal-driven, implementation complexity

The Long-Term Pattern

Voice = Mature, saturated, incremental growth in emerging markets

Non-Voice = Exponential, compounding, unreplicable organizational IP, permanent cost structure transformation, builds institutional knowledge, creates switching costs

Mapping the vertical AI landscape

INDUSTRY	NAICS CODE	KEY WORKFLOW AUTOMATION	2025 SPEND	2030 TAM	CAGR	ADOPTION
Retail & E-commerce	44-45	Personalized recommendations, inventory management, dynamic pricing, customer support, loss prevention, order fulfillment	$46.74B	$175.11B	30.2%	MEDIUM-HJIGH
Financial Services & Banking	52	Fraud detection, loan processing, KYC/AML compliance, invoice processing, treasury operations, contract analysis, claims processing	$5.51B	$33.26B	40.8%	HIGH
Manufacturing & Industrial	31-33	Production planning, predictive maintenance, supply chain optimization, quality control, inventory management, demand forecasting	$8.67B	$13.50B	14.2%	MEDIUM
Telecommunications & Media	51,517	Network fault detection, 5G optimization, customer service, billing management, technical support, network configuration	$2.50B	$8.50B	28.0%	MEDIUM
Government & Public Sector	92	Grant management, healthcare administration, benefits coordination, security threat detection, policy analysis, case management	$1.20B	$5.20B	35.0%	MEDIUM
Healthcare & Life Sciences	62	Clinical documentation, patient monitoring, diagnostic support, claims processing, resource allocation, staff scheduling, administrative tasks	$0.54B	$4.96B	45.6%	MEDIUM-HIGH
Legal & Professional Services	54	Contract analysis, legal research, document review, due diligence, compliance monitoring, deposition preparation	$1.00B	$4.50B	35.5%	MEDIUM
Energy, Utilities & Resources	22,21	Grid management, demand forecasting, outage detection, predictive maintenance, asset management, regulatory reporting	$0.80B	$2.80B	28.5%	MEDIUM

Quant Sizing

AI & Finance

KEY WORKFLOWS

$0.7T – $1.0T

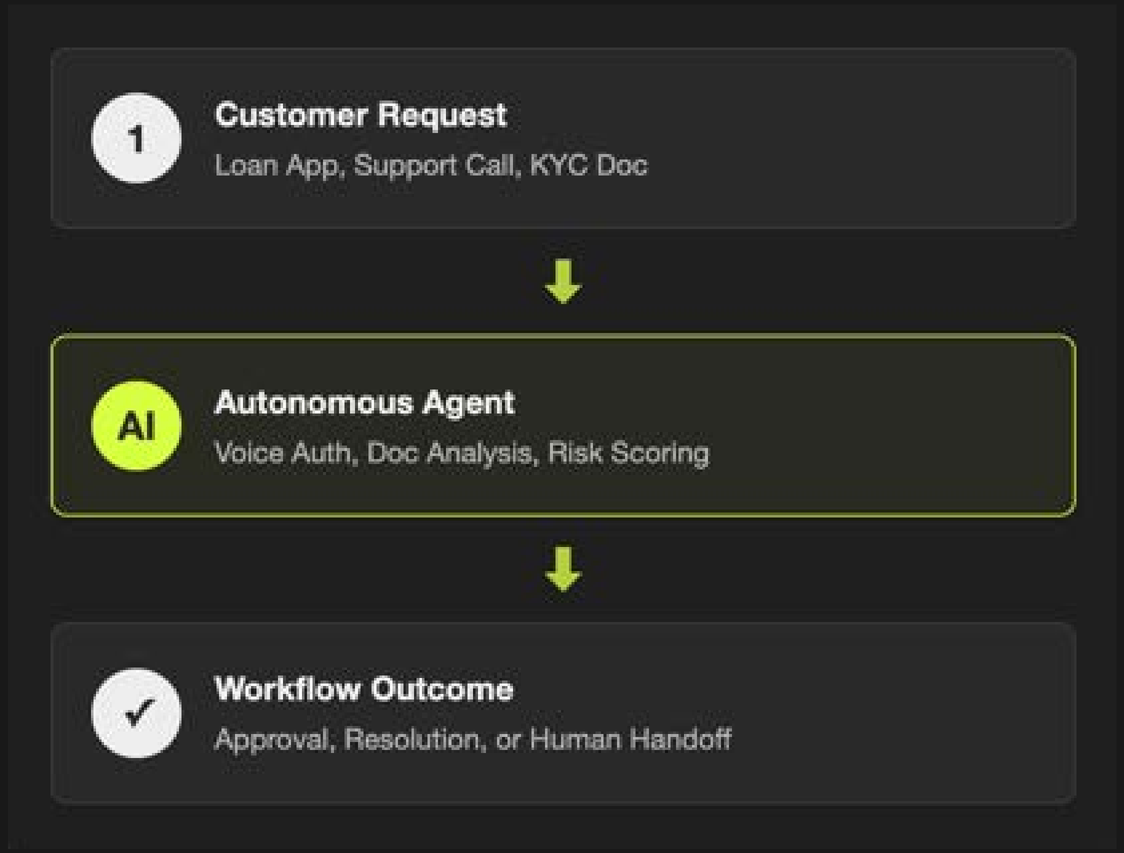
TOTAL ADDRESSABLE MARKET (TAM)

Total economic value unlocked via AI agent automation across global BFSI.

$11B – $34B

SERVICEABLE ADDRESSABLE MARKET (SAM)

Annual vendor market capture based on current spending trajectories.



/1 **Customer Onboarding & KYC:** Automated document verification and identity checks reducing manual review by 60%.

/2 **Loan Origination:** End-to-end approval workflows for retail products.

/3 **Fraud Detection:** Real-time transaction monitoring and compliance reporting.

/4 **Customer Support & Collections:** Empathetic, compliant voice agents for high-volume outreach.

Source: McKinsey 2024; Gartner Conversational AI Forecast 2025; IDC Spending Guide 2025

TAM = Total Banking AI Value × Share of workflows captured by agents
SAM = TAM × % of market accessible to third-party vendors

Adoption Status: Medium-High among Tier 1 banks; Low-Medium for mid-market regional banks.

Quant Sizing

AI & Finance

FINANCIAL SERVICES & INSURANCE SENSITIVITY ANALYSIS SCENARIOS

- Gen AI in banking could unlock $200-340 bn/yr (~2.8-4.7% of revenues)
- Agentic AI may resolve ~80% of routine service issues by 2029, reducing cost by ~30%
- >40% of agentic AI projects are predicted to be cancelled by 2027 due to governance/ROI issues
- Global AI spend expected to grow to $631bn by 2028 at ~30% CAGR

SCENARIO	ENTERPRISE ADOPTION (%)	AVG SPEND / FIRM ($M)	AI WORKFLOWS CAPTURED	ESTIMATED TAM ($B)	ESTIMATED SAM ($B)
Conservative	20%	$1.0M	5	$28B	$8B
Base Case	35%	$2.5M	8	$60B	$18B
Aggressive	50%	$4.0M	10	$95B	$28B

Insight: Base case assumes moderate regulatory headwinds. Aggressive scenario unlocked if regulatory bodies (e.g., OCC, FCA) provide clear frameworks for autonomous agent compliance by mid-2026.

Source: IDC 2025; McKinsey 2024

LAYER	COMPANY	CORE FOCUS	KEY DIFFERENTIATION
Application	**Kasisto**	Conversational Banking	Proven large-scale deployment (JPM, TD Bank); KAI platform deeply trained on financial intent.
	Clinc	AI Digital Assistants	Proprietary NLP stack specifically fine-tuned for messy, complex banking queries.
Infrastructure	**Cognigy**	Multi-channel Orchestration	Low-code integration with legacy core banking APIs (Fiserv, FIS).
	Gupshup	Bot Management Platform	Scalable pipelines for managing thousands of concurrent agent interactions in emerging markets.
Services	**Accenture**	Implementation & Compliance	Deep BFSI partnerships; bridge between legacy compliance and new AI agent risk frameworks.
	Infosys	AI Rollout & Integration	Global delivery model for rapid scaling of proven agent workflows.

Source: a16z "AI Agents Report", 2025

Quant Sizing

Retail & E-commerce

KEY WORKFLOWS

$3T – $5T

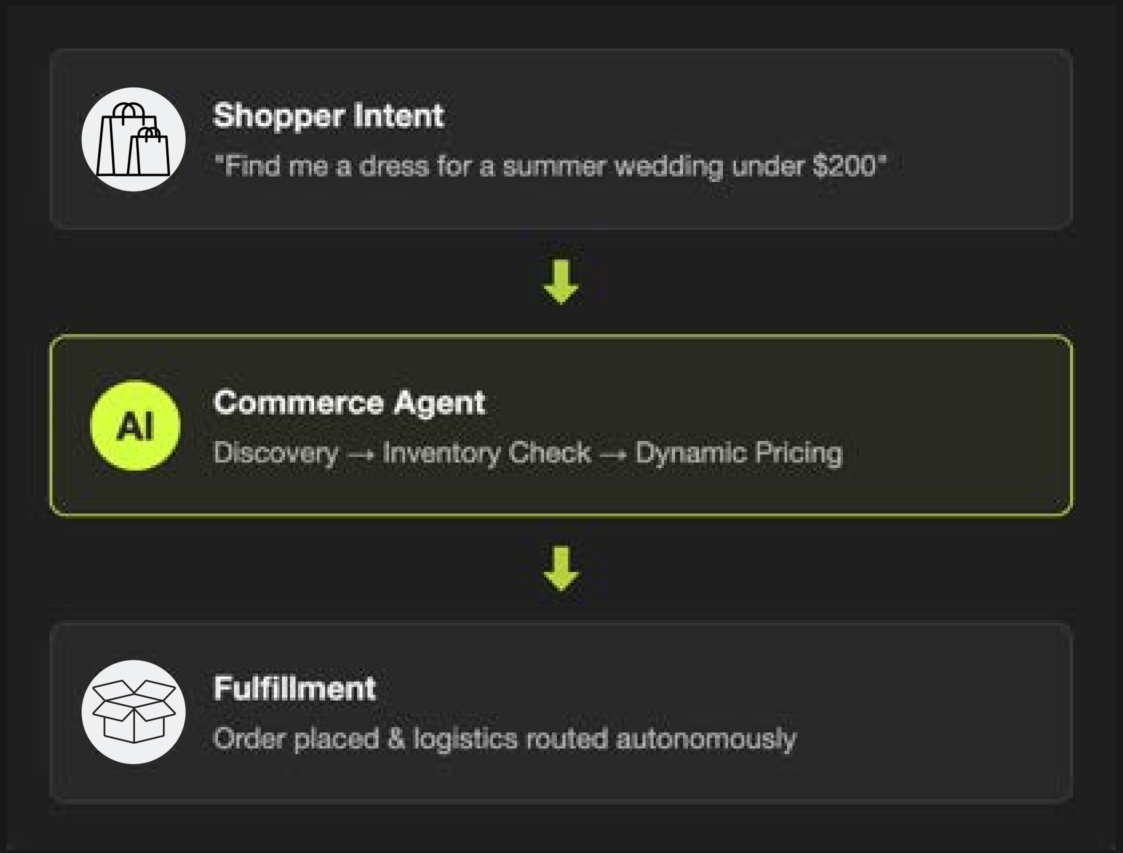
GLOBAL TAM (BY 2030)

Total value of retail workflows addressable by agentic automation globally.

$7.5B – $25B

ANNUAL SAM

Immediate addressable vendor market for retail AI agents.



/1 **Conversational Commerce: Voice/Chat agents that guide discovery to checkout.**

/2 **Personalized Recommendations:** Real-time, context-aware agentic suggestions.

/3 **Returns & Fulfillment:** Autonomous processing of returns and logistics routing.

/4 **Dynamic Pricing & Inventory:** Agents adjusting to real-time demand signals.

Source: McKinsey "The Agentic Commerce Opportunity", 2025; Gartner Retail AI Forecast, 2025

TAM = Total Banking AI Value × Share of workflows captured by agents
SAM = TAM × % of market accessible to third-party vendors

Adoption Status: High in top-tier platforms (Amazon, Shopify Plus merchants); rapidly emerging in mid-tier direct-to-consumer.

RETAIL SENSITIVITY ANALYSIS SCENARIOS

- Retailers with deep AI workflow integration (e.g., personalization, returns, supply chain) can achieve 15-20% cost reductions and 5-8% revenue increases (McKinsey).
- By 2025, only about 11% of U.S. retailers feel fully ready for large-scale AI deployment (Amperity survey), highlighting the importance of readiness.
- A significant number of apps will incorporate agentic AI by 2028 (Gartner), suggesting a shift from limited to broader workflow automation, enhancing segment potential.

SCENARIO	RETAILERS USING AGENTS (%)	AVG SPEND / RETAILER ($M)	KEY WORKFLOWS	TAM ($B)	SAM ($B)
Conservative	15%	$0.6M	4	$18B	$5B
Base Case	30%	$1.2M	6	$42B	$12B
Aggressive	45%	$2.5M	8	$70B	$22B

Insight: Aggressive scenario heavily dependent on success of voice commerce adoption in mobile apps, moving beyond just text-based chatbots.

Source: McKinsey 2025; Gartner 2025; IDC 2025

Key Landscape

Retail & E-commerce

LAYER WISE CLASSIFICATION

LAYER	COMPANY	CORE FOCUS	KEY DIFFERENTIATION
Application	**Gorgias**	Conversational Support	Deep Shopify integration; high conversion rates turning support tickets into sales opportunities.
	Sierra AI	Workflow Automation	Agentic order processing that handles complex, multi-turn returns and exchanges autonomously.
Infrastructure	**Beam AI**	Multi-agent Orchestration	Framework supporting cross-merchant integration, vital for marketplaces and aggregators.
	Cognigy	Conversational Platform	Enterprise-grade scalable deployment suitable for global omnichannel retailers.
Services	**Deloitte**	Implementation & Analytics	Expertise in tying agent performance directly to omnichannel revenue metrics.
	PwC	Enterprise Retail AI Rollout	Global systems integration for large-scale legacy retail transformation.

Source: a16z "AI Agents Report", 2025

Strategy

Paucek and Lage

2024

INVESTMENT THESIS & GROWTH FLYWHEEL

1. ENTRY WEDGE
Automate high-volume, low-risk tasks (Customer Support, basic Loan Origination).

2. EXPANSION
Move up-stack to complex multi-agent systems (Financial Concierge, Wealth Advisory).

3. DEFENSIBILITY
Lock-in via proprietary compliance workflows & voice auth data.

4. SCALE
Cross-sell across retail, commercial, and wealth management divisions.

invest in platforms that leverage low-risk entry wedges to build proprietary data moats in high-stakes financial workflows.

Primary Risk: Regulatory clampdown on autonomous financial advice.
Key Catalyst: Open Banking API standardization accelerating agent capabilities.

Fin Services & Inv.

Back infrastructure that transforms static e-commerce into dynamic, **agent-driven marketplaces**, starting with high-friction post-purchase flows.

Primary Risk: Commoditization of basic support agents by major platforms (Shopify/Amazon native tools)
Key Catalyst: Widespread adoption of voice-first shopping interfaces (smart glasses, next-gen assistants)

1. ENTRY WEDGE
Conversational commerce & automated returns processing to prove immediate ROI.

2. EXPANSION
Supply chain orchestration and autonomous agentic fulfillment.

3. DEFENSIBILITY
Platform play with multi-merchant data network effects & voice checkout IP.

4. SCALE
Standard for all merchant interactions; B2B and B2C unification.

Retail & E-comm

Source: McKinsey 2024; Deloitte 2025

CUSTOMER CENTRICITY

Customer interaction complexity = financial transaction complexity, the differentiation is compliance risk, not domain difficulty

- 85% customer service mkt adoption vs. 60% of finance delayed
- 65-70% success rate vs. only 38% in finance
- Risk-adjusted ROI converges ($35B-$56B net)
- Compliance spend explodes in finance ($79B opportunity cost)

Strategic Pathway: Customer service → sales/retention/collections → financial services (de-risked with proven playbook)

Deloitte Financial AI Adoption Report 2024, Gartner 2025 Customer Service Survey

Financial services isn't Plan B—it's Plan C. Customer service is the profitable, defensible wedge that de-risks it.

Moat	Mechanism	Switching Cost
Data Moat	Billions of interactions train proprietary NLU; competitors need 5 years to replicate	20-30% accuracy disadvantage
Workflow Lock-In	Customer's escalation logic, tone, FAQ = embedded IP; ripping out = operational chaos	6-12 months migration
Infrastructure Switching	Deep Salesforce/Genesys/Zendesk integration; re-integration = $500K+	Contact center goes dark during migration
Partnership Moat	White-labeled agents sold by contact center platforms	Competitors shut out of distribution
Compliance Credibility	Audit-ready governance proven in customer service → faster financial services deployment	40% faster regulatory approval

ENTRY WEDGE: BILLING & ACCOUNT SUPPORT

- ROI: Billing inquiries are 20-30% of all contact center volume with proven 60-80% deflection rates
- 12x cost advantage: $0.50 AI vs. $6.00 human agent cost per interaction
- Repeatable expansion: Billing → FAQ → Scheduling → Sales (each reuses core infrastructure)
- Zero regulatory friction: Non-transactional, no approvals/denials, fits existing workflows
- Data moat foundation: Millions of interactions train your proprietary NLU models

Gartner 2025 Customer Service Survey, Fullview 2025 Customer Service AI Statistics , Salesforce AgentForce ROI Data, BCG Agentic Commerce Report, Deloitte Financial AI Adoption Report 2024, McKinsey Global AI Survey 2025

Projection

Cozmo AI Assgn

11/10/2025

AGENTIC AI TIMELINE

2026

Launch an integrated omnichannel AI layer to manage voice, chat, SMS, and social interactions from a single dynamic workflow, ensuring no silos, instant context, and continuous learning.

Mid 2027

Autonomous Resolution with Proactive Personalization: Move to proactive AI where agents autonomously address over 80% of customer needs and engage users with reminders, status updates, and offers before they reach out.

2028

Human-Level Emotional Intelligence & Compliance-Grade Decisioning: AI agents possess advanced emotional intelligence (EQ) to detect nuances and sentiment, enabling empathetic engagement. Ensure compliance by maintaining complete, auditable logs of all interactions.

Mid 2029

Trusted Autonomous Orchestrator for Revenue & Retention: By Year 3, customer service AI platforms will autonomously handle retention, upselling, cross-selling, and complex exceptions, acting as a continuous, revenue-driving relationship manager.